

**Project 648-20-121 -
Proposed Campus
Security Fence**

**Portland VAMC
Vancouver Campus
Vancouver, Washington**

**Draft Report
Geotechnical Subsurface Investigation**

**Blue Trident, LLC
Bainbridge Island,
Washington**

October 24, 2022

TTL Project No. 2213101



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October 24, 2022

TTL Project No. 2213101

Mr. Brendan Geraghty
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Draft Report
Geotechnical Subsurface Investigation
Project 648-20-121 – Proposed Campus Security Fence
Portland VA Medical Center – Vancouver Campus
Vancouver, Washington

Dear Mr. Geraghty:

Following is the report of the geotechnical subsurface investigation performed by TTL Associates, Inc. (TTL) for the referenced project. This study was performed in accordance with TTL Proposal No. 2213101, dated March 2, 2022 and was authorized with Blue Trident, LLC Addendum No. 001 to Subcontract No. 31-18-0001, dated July 5, 2022.

This report contains the results of our study, our engineering interpretation of the results with respect to the project characteristics, and our recommendations for design and construction of campus security fence foundations. In accordance with U.S. Department of Veterans Administration (VA) protocol, this report is being submitted as “Draft” pending questions and comments by Blue Trident, LLC and VA. However, the report is considered complete and comprehensive with respect to the requested scope of work.

Soil samples collected during this investigation will be stored at our laboratory for 90 days from the date of this report. The samples will be discarded after this time unless you request that they be saved or delivered to you.

Should you have any questions regarding this report or require additional information, please contact our office.

Sincerely,

TTL Associates, Inc.

Christopher P. Iott
Geotechnical Project Manager

Curtis E. Roupe, P.E.
Vice President

c.c.: Mr. Brian E. Moll – Blue Trident, LLC

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**DRAFT REPORT
GEOTECHNICAL SUBSURFACE INVESTIGATION
PROJECT 648-20-121 – PROPOSED CAMPUS SECURITY FENCE
PORTLAND VA MEDICAL CENTER – VANCOUVER CAMPUS
VANCOUVER, WASHINGTON**

FOR

**BLUE TRIDENT, LLC
900 WINSLOW WAY E, SUITE 130
BAINBRIDGE ISLAND, WASHINGTON 98110**

SUBMITTED

**OCTOBER 24, 2022
TTL PROJECT NO. 2213101**

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1.0 INTRODUCTION

This geotechnical subsurface investigation report has been prepared for the proposed replacement with a new campus security fence of the existing 6 feet high chain-link fence around the Portland VA Medical Center – Vancouver Campus, located at 1601 East 4th Plain Boulevard, in Vancouver, Washington. The general area of the project is shown on the attached Site Location Map (Plate 1.0).

This report summarizes our understanding of the proposed construction, describes the investigative and testing procedures, presents the findings, discusses our evaluations and conclusions, and provides our design and construction recommendations for fence support foundations.

This study was performed in accordance with TTL Proposal No. 2213101, dated March 2, 2022 and was authorized with Blue Trident, LLC Addendum No. 001 to Subcontract No. 31-18-0001, dated July 5, 2022.

The purpose of this investigation was to evaluate the subsurface conditions and laboratory data relative to the design and construction of fence support foundations at the referenced location. This investigation included eight test borings, field and laboratory soil testing, and a geotechnical engineering evaluation of the test results.

This report includes:

- A description of the subsurface soil and groundwater conditions encountered in the borings.
- Design recommendations for fence post foundations related to the proposed development.
- Recommendations concerning soil- and groundwater-related construction procedures such as site preparation, earthwork, foundation construction, and related field testing.

The scope of this investigation did not include an environmental evaluation of the site or subsurface conditions.

2.0 INVESTIGATIVE PROCEDURES

This subsurface investigation included eight test borings, designated as B-1 through B-8, drilled on September 22 and 23, 2022. The borings were performed by Cascade under the direction of TTL. The borings were performed at approximate equal spacing adjacent to the existing chain link fence around the VA Medical Center campus. The test borings were located in the field by Cascade based on a proposed plan that was approved by Mr. Andrew S. Thornton of Portland VA Medical Center, Vancouver Campus on August 10, 2022. Ground surface elevations at the boring locations were estimated using Google Earth. The approximate locations of the borings are shown on the Test Boring Location Plan (Plate 2.0).

The test borings performed during this investigation were advanced with a GeoProbe® 3230DT utilizing 4-inch casing. All borings were terminated at depth 16½ feet below existing grade.

During borehole advancement, soil samples were collected at 2½-foot intervals to a depth of 10 feet and at 5-foot intervals thereafter. Split-spoon (SS) samples were obtained by the Standard Penetration Test (SPT) Method (ASTM D 1586), which consists of driving a 2-inch outside diameter split-barrel sampler into the soil with a 140-pound weight falling freely through a distance of 30 inches. The sampler was driven in three successive 6-inch increments with the number of blows per increment being recorded. The sum of the number of blows required to advance the sampler the second and third 6-inch increments is termed the Standard Penetration Resistance (N-value) and is presented on the Logs of Test Borings attached to this report. The samples were sealed in jars and shipped to our laboratory for further classification and testing.

All of the recovered samples of the subsoils were visually or manually classified in accordance with the Unified Soil Classification System (USCS) (ASTM D 2487 and D 2488). All of the recovered samples were also tested in our laboratory for moisture content (ASTM D 2216). For corrosivity considerations, three selected samples were tested for pH (ASTM D 4972). Atterberg Limits tests (ASTM D 4318) and particle size analyses (ASTM D 6913 and D 7928) were performed on samples from Borings B-1 (SS-1), B-3 (SS-1), and B-5 (SS-1). Additionally, particle size analyses for granular soil samples were performed on samples from Borings B-6 (SS-4) and B-8 (SS-4) to determine soil classification properties. The results of these tests are presented on the Logs of Test Borings and Grain Size Distribution sheet attached to this report.

Soil conditions encountered in the test borings are presented in the Logs of Test Borings, along with information related to sample data, SPT results, water conditions observed in the borings, and laboratory test data. It should be noted that these logs have been prepared on the basis of laboratory classification and testing as well as field logs of the encountered soils.

Experience indicates that the actual subsoil conditions at a site could vary from those generalized on the basis of test borings made at specific locations, especially at previously developed sites such as this site. Therefore, it is essential that a geotechnical engineer be retained to provide soil engineering services during the site preparation, excavation, and foundation phases of the proposed project. This is to observe compliance with the design concepts, specifications, and recommendations, and to allow design changes in the event subsurface conditions differ from those anticipated prior to the start of construction.

3.0 PROPOSED CONSTRUCTION

The proposed construction consists of replacement with a new campus security fence of the existing 6 feet high chain-link fence around the Portland VA Medical Center – Vancouver Campus, located at 1601 East 4th Plain Boulevard, in Vancouver, Washington. The fence will be approximately 1.1 miles long (approximately 5,800 feet). A concrete mow strip along with two motorized vehicle gates and two personnel gates is included with the installation of this new security fence. Structural loads were not provided, but are assumed to be light in magnitude. Maximum column loads are assumed to not exceed 40 kips. Smaller steel fence post type foundation loads are assumed to be on the order of 200 to 400 pounds, although uplift or lateral loads may govern embedment requirements.

We have assumed that final grades will approximate existing grades present at the time of this investigation.

4.0 GENERAL SITE AND SUBSURFACE CONDITIONS

4.1 General Site Conditions

The perimeter fence line of the VA Medical Center campus included grass areas and adjacent parking lots. Approximate ground surface elevations obtained from Google Earth for the borings locations varied from Elevs. 211± to 171±, with the higher elevations present in the north and the lowest elevations present in the southwest.

The borings were generally performed in grass areas and encountered surface materials consisting of topsoil, with thickness on the order of 3 inches. Borings B-3 and B-7 were performed in pavement areas and encountered approximately 6 inches of asphalt at each location. Distinct surface cover was not present at the location of Boring B-6.

4.2 General Soil Conditions

The subsoils encountered to the depth of termination of 16 ½ feet below existing grade consisted of predominantly **loose** to medium dense granular soils. The granular soils consisted of silty sand (SM), poorly graded sand with varying amounts of silt (SP and SP-SM), as well as clayey sand, each with varying amounts of gravel. SPT N-values generally ranged from 5 to 18 blows per foot (bpf). Moisture contents ranged from approximately 7 to 16 percent.

Within the upper 5 feet of Borings B-3, B-6, and B-7, predominantly medium stiff to stiff sandy silt (ML) with trace gravel was encountered. The tested sample from Boring B-3 contained 51 percent silt and clay fraction, just above the threshold of 50 percent between silty sand (SM) and sandy silt (ML) designation. As such, the sandy silt soils may be difficult to distinguish from the silty sand soils at the site. In any case, each soil type is generally suitable for support of the foundations associated with the proposed new campus security fence. Within the sandy silt zones, SPT N-values of 5 bpf, 14 bpf, and 23 bpf were determined. An unconfined compressive strength of 2,000 pounds per square foot (psf) was determined for the lone intact sample, which was obtained from Boring B-3. Moisture contents ranged from 10 to 17 percent.

Additional descriptions of the stratigraphy encountered in the borings are presented on the Logs of Test Borings.

4.3 Groundwater Conditions

Groundwater was initially encountered during drilling in only Boring B-1 at a depth of 15 feet below existing grade (Elev. 194±). Groundwater was not encountered in the remaining borings and was not observed upon completion of drilling in any of the borings. It should be noted that each of the borings was performed and backfilled within the same day, and stabilized water levels may not have occurred over this limited period. Instrumentation was not installed to observe long-term groundwater levels.

Based on the limited data available, such as the soil characteristics and the groundwater conditions encountered in the borings, it is our opinion that the “normal” groundwater level may be encountered at depths below the extents of the borings performed for this investigation. However, this investigation did not include research of possible hydrological influences at the project site. It should be noted that groundwater elevations can fluctuate with seasonal and climatic influences. Therefore, the groundwater conditions may vary at different times of the year from those encountered during this investigation.

5.0 DESIGN RECOMMENDATIONS

The following conclusions and recommendations are based on our understanding of the proposed construction and on the data obtained during the field investigation. If the project information or location as outlined is incorrect or should change significantly, a review of these recommendations should be made by TTL. These recommendations are subject to the satisfactory completion of the recommended site and subgrade preparation and fill placement operations described in Section 6.0, “Construction Recommendations”.

5.1 Shallow Spread Foundations

This section includes campus security fence foundation recommendations considering shallow spread foundations. If fence-post type foundations will be utilized, recommendations are provided in the following section.

It is recommended that spread foundations bear at or below a minimum depth of 12 inches below finished grades. Based on the results of the field and laboratory testing for the borings performed for this investigation, spread foundation excavations are anticipated to encounter predominantly **loose** to medium dense native granular soils, as well as zones of predominantly medium stiff to stiff native cohesive soils. These soils are considered generally suitable for the foundations, although in-place densification of loose granular soils will be required at the bottom of the footing excavation. Where loose granular soils are present at the bearing elevation, they should be re-compacted in-place using a backhoe-mounted vibratory compactor (hoe-pac) or similar equipment, or removed and replaced with new granular engineered fill.

Suitable compaction/bearing of native granular foundation soils can be verified as:

- Exhibiting a compacted (in-situ) dry density of at least 100 percent of the maximum dry density determined by Standard Proctor (ASTM D 698) laboratory compaction,
- A dynamic cone penetrometer (DCP) reading of at least 8 blows per increment (average over three increments), or
- Other methods to demonstrate an equivalent SPT N-value of 10 bpf or greater.

Following the satisfactory completion of the site preparation and footing excavation inspections outlined in this report, the proposed structure may be supported on a conventional shallow foundation system. For individual (square) footings, we recommend a net allowable bearing pressure of 2,000 pounds per square foot (psf), with a maximum bearing pressure at the toe of 2,665 psf due to overturning (if applicable, such as with wind load considerations).

In using a net allowable soil pressure, the weight of the footings or backfill over the footings need not be included in the structural loads for dimensioning footings. A friction factor of 0.35 may be utilized along the base of the footing to calculate sliding resistance. The bearing materials should be field-verified as being native granular soils meeting the compaction/bearing criteria presented above, native cohesive soils with an unconfined compressive strength of at least 2,000 psf, or properly placed and compacted new engineered fill.

Due to the encountered very loose to loose granular soils as well as the existing site development, it is particularly important that a detailed foundation excavation inspection be completed by a geotechnical engineer or qualified representative. Inspection should be performed to verify that the exposed soil conditions at the bearing elevations are consistent with the subsurface conditions encountered in the test borings, loose granular soils have been suitably modified in-place, and that the conditions are suitable for foundation bearing. Additionally, the presence of our engineer will help facilitate the timely remediation of unsuitable soil conditions. If the results of DCP, hand penetrometer, or other strength tests indicate the exposed soil conditions are not suitable for the design bearing pressure, it may be necessary to increase the footing size to accommodate the lower bearing strengths or to over-excavate and backfill with engineered fill or flowable fill.

If loose native granular soils are encountered that cannot be suitably re-compacted in-place, or if other deleterious materials are encountered, over-excavation should extend through these materials to suitable bearing soils. The base of the over-excavation should be widened one foot for every foot of depth below the planned bearing depth, centered along the footing. The over-excavated areas should be backfilled with dense-graded aggregate, placed in controlled lifts, and compacted to not less than 100 percent of the maximum dry density as determined by ASTM D 698 (Standard Proctor). Alternatively, the over-excavated areas could be backfilled with lean concrete having a minimum compressive strength of 1,500 pounds per square inch (psi) or other flowable controlled-density fill having a minimum compressive strength of 300 psi. If foundations will be placed at the base of the over-excavation or the lean concrete fill option will be utilized, widening the footing over-excavation will not be required. If the controlled-density fill option is utilized, the footing over-excavation shall be widened as discussed above.

All exterior footings and footings in unheated areas should be constructed at a minimum depth of 12 inches below finished exterior grades. Column (square) footings should be at least 30 inches wide, regardless of sizing based on design loads and the recommended allowable bearing pressure. It should be noted that use of trench footings (i.e., placement of foundation

concrete without forming) is not expected to be feasible due to the granular nature of the bearing soils, along with the need for in-place densification. Forming of footings and backfilling around CMU or poured foundation walls should be anticipated.

Utilizing the above allowable bearing pressure and proper foundation inspection techniques, the total settlement associated with the structure should not exceed 1 inch and differential settlement should not exceed $\frac{3}{4}$ inch.

5.2 Post Foundations

This section includes campus security fence foundation recommendations considering steel fence-post type foundations. If shallow spread foundations will be utilized, recommendations are provided in the preceding section.

5.2.1 Vertical Capacity

The steel fence post-type foundations are anticipated to bear in predominantly **loose** to medium dense native granular soils. Although zones of sandy silt were encountered in the upper 5 feet of three borings, vertical capacity design parameters based on the granular soils would dictate design. Additionally, a tested sample of the encountered sandy silt soils contained only 51 percent silt and clay fraction. As such, the vertical capacity provided by these soils is anticipated to be similar to the capacity provided by the silty sand soils at the site.

Vertical downward load and uplift load may be resisted by the side resistance on the embedded pipe. However, due to freeze-thaw effects and moisture content changes in the upper profile, effective side resistance may not be reliable in the upper 12 inches. **Therefore, we recommend that side resistance be considered only for the embedment zone below a depth of 12 inches below final grade.**

It should be noted that a minimum factor of safety (FoS) of 3 is recommended when calculating allowable capacity for pile-type foundations due to the variability of soil strength and installation methods, without performance of load tests. Based on this FoS, we recommend design (allowable) side resistance of 35 pounds per square foot (psf) for vertical compression and 22 psf for vertical uplift considerations.

A FoS of 2 may be utilized for design if load tests are performed. Load tests should be performed in accordance with ASTM D 1143, "Standard Test Method for Piles Under Static Axial Compressive Load" and ASTM D 3689, "Standard Test Method Individual Piles Under

Static Axial Tensile Load.” Based on utilization of load tests, and a FoS of 2 applied to calculated ultimate values, we recommend design (allowable) side resistance of 55 psf for vertical compression and 32 psf for vertical uplift considerations.

5.2.2 Lateral Capacity

For lateral load-deflection evaluations using software, such as LPILE, recommended design parameters are summarized in the following table based on the conditions encountered in the borings.

Table 5.2.2.A. Subsurface Conditions and Recommended Lateral Load-Deflection Parameters				
General Condition				
Average Depth (feet)	Generalized Layer Description	Approximate Total Unit Weight (pcf)	Average Internal Angle of Friction, ϕ (degrees)	Soil-Modulus Parameter, k (pci)
0 to 16½	Loose to Medium Dense Granular Soils	120	30	25

In addition to lateral load-deflection evaluation using the conditions presented in Table. 5.2.2.A, for steel post foundations in the areas of Borings B-3, B-6, and B-7, consideration should also be given to the upper profile sandy silt soils in these three borings using the design values in the following table. In these areas, design should be based on the more conservative result modeled by the conditions in Tables 5.2.2.A and 5.2.2.B.

Table 5.2.2.B. Subsurface Conditions and Recommended Lateral Load-Deflection Parameters						
Sandy Silt over Silty Sand Condition (B-3, B-6, and B-7)						
Average Depth (feet)	Generalized Layer Description	Approximate Total Unit Weight (pcf)	Average Undrained Shear Strength, S_u (psf)	Strain at 50% Maximum Stress, ϵ_{50}	Average Internal Angle of Friction, ϕ (degrees)	Soil-Modulus Parameter, k (pci)
0 to 5	Medium Stiff to Stiff Cohesive Soils	120	1,100	0.007	-	-
5 to 16½	Loose to Medium Dense Granular Soils	120	-	-	30	25

5.3 Seismic Considerations

We have reviewed seismic design parameters in accordance with International Building Code (IBC) criteria. It should be noted that the IBC seismic site characterization is based on the upper 100 feet of the geologic profile and the borings performed for this investigation extended only to a maximum depth of 16½ feet below existing grade. Therefore, our analysis is limited to characterization based on the encountered overburden soil profile only.

Based on IBC Section 1613.3.2, which references ASCE 7-10, using the N-method, the weighted average N-value for the profile was calculated to be less than 15 blows per foot (bpf). Therefore, the site can be characterized as Site Class E “soft soil” in accordance with ASCE 7-10 Table 20.3-1.

If a Site Class E designation will be restrictive to structural design, it may be prudent to perform deeper SPT borings for seismic site class evaluations, to determine if a better site class designation is appropriate. However, there is no guarantee that such testing would yield a more favorable site class designation.

An alternate method to evaluate seismic site class incorporates shear wave velocity. Development of a shear wave velocity profile is typically performed in conjunction with down-hole or cross-hole seismic testing, which would require additional boreholes and specialized testing equipment, cone penetration testing (CPT) using a CPT rig, or Multi-channel Analysis of Surface Waves (MASW) testing. It should be noted that, like the deeper SPT borings, there is no guarantee that shear wave velocity testing would yield a more favorable site class designation.

5.4 Corrosion Considerations

Soil samples from Borings B-2, B-4, B-6, and B-8 were tested for pH. The pH results are summarized as follows:

Table 5.4. Corrosion Considerations Test Results		
Boring No. (Sample No.)	Soil Type	pH
B-2 (SS-4)	Poorly Graded Sand (SP)	6.8
B-4 (SS-2)	Silty Sand (SM)	7.2
B-6 (SS-1)	Sandy Silt (ML)	7.4
B-8 (SS-1)	Poorly Graded Sand w/Silt (SP-SM)	6.9

This range of pH is characterized as neutral to slightly alkaline soil reaction by the USDA Soil Conservation Service. Typically, soils with a pH range between 5 and 9 are not considered to represent a significant corrosion risk to buried structural concrete or underground utilities. Therefore, we do not expect that these soils will represent a significant corrosion risk to underground construction, although this assessment is based on limited data.

5.5 Groundwater Control and Drainage

As stated previously, groundwater was initially encountered during drilling in only Boring B-1 at a depth of 15 feet below existing grade (Elev. 194±). Groundwater was not encountered in the remaining borings and was not observed upon completion of drilling in any of the borings. Based on the limited data available, such as the soil characteristics and the groundwater conditions encountered in the borings, it is our opinion that the “normal” groundwater level may be encountered at depths below the extents of the borings performed for this investigation.

It is our experience that adequate control of groundwater seepage or surface water run-off into shallow excavations should be achievable by minor dewatering systems, such as pumping from prepared sumps. Although not anticipated to be required, if deeper excavations in granular soils were to extend below the groundwater table, installation of well points would likely be required in addition to pumping from prepared sumps. The type of dewatering system utilized will depend on construction practices, soil conditions encountered in the foundation excavations, seasonal conditions, and the depth of excavation. Additionally, the contractor will need to exercise diligence to control seepage and runoff to maintain a stable subgrade.

In the event excessive seepage is encountered during construction, TTL may be notified to evaluate whether other dewatering methods are required.

5.6 Excavations and Slopes

The sides of temporary excavations for foundations, utility installations, and other construction should be adequately sloped to provide stable sides and safe working conditions. Otherwise, the excavation must be properly braced against lateral movements. In any case, applicable Occupational Safety and Health Administration (OSHA) safety standards must be followed.

The soils encountered during this investigation, within the anticipated depths of excavation, consist of the following OSHA Type soils:

- OSHA Type A soils (cohesive soils with unconfined compressive strengths of 3,000 pounds per square foot (psf) or greater),
- OSHA Type B soils (cohesive soils with unconfined compressive strengths greater than 1,000 psf but less than 3,000 psf), and
- OSHA Type C soils (granular soils).

For temporary excavations in Type A, B, and C soils, side slopes must be no steeper than $\frac{3}{4}$ horizontal to 1 vertical ($\frac{3}{4}$ H:1V), 1H:1V, and 1½H:1V, respectively. For situations where a higher strength soil is underlain by a lower strength soil and the excavation extends into the lower strength soil, the slope of the entire excavation is governed by that required for the lower strength soil. In all cases, flatter slopes may be required if lower strength soils or adverse seepage conditions are encountered during construction.

For permanent excavation slopes, we recommend that grades be no steeper than 3H:1V without a more extensive geotechnical evaluation of the proposed construction plans and site conditions.

6.0 CONSTRUCTION RECOMMENDATIONS

6.1 Site and Subgrade Preparation

Prior to proceeding with construction operations, all topsoil, root mat, vegetation, pavements, and other deleterious non-soil materials should be removed from the proposed construction areas. Suitable topsoil may be stockpiled for later use in landscaped areas. It is important to note that topsoil thicknesses referenced in the borings may vary across the site. Typically, soils with more than 5 percent organics are not recommended as subgrade soils in structure areas, but dark colored soils having the appearance of topsoil with only trace “root hairs” of 5 percent or less may not necessarily require stripping. For these “transitional” soils, the actual moisture content and subgrade stability under proof-rolling operations is more critical than the color in determination of the amount of stripping or subgrade undercut. The actual amount of required stripping should be determined in the field by a geotechnical engineer or qualified representative.

Upon completion of stripping and clearing, the areas intended to support new fill and foundations should be carefully inspected by a geotechnical engineer.

6.2 Fill

Material for engineered fill or backfill required to achieve design grades may consist of any non-organic soils having a maximum dry density as determined by the Standard Proctor (ASTM D 698) of 90 pounds per cubic foot (pcf) or greater. On-site soils may be re-used as engineered fill materials provided that they contain trace or less organic matter and debris, and are free of excessive moisture, and rock or stone fragments larger than 3 inches in diameter. Depending on seasonal conditions, the on-site soils may be wet of optimum and may require scarification and aeration to achieve satisfactory compaction. If the construction schedule does not allow for scarification and aeration activities, it may be more practical or economical to utilize imported granular fill.

Fill should be placed in uniform layers no more than 8 inches thick (loose measure) and adequately keyed into stripped and scarified soils. All fill within the structure areas should be compacted to not less than 100 percent of the maximum dry density as determined by ASTM D 698 (Standard Proctor).

The upper soil profile at the site consists of native granular soils, as well as native sandy silt soils. The contractor should be prepared to use a sheepsfoot roller to provide effective compaction of the sandy silt soils, as well as particularly silty and clayey granular soils. The

contractor should be prepared to use a vibratory, smooth-drum roller to provide effective compaction of the silty and clayey granular soils that do not compact well with a sheepfoot roller, for granular soils containing less silt and clay fraction (SP or SP/SM), as well as for new granular engineered fill. In narrow utility or footing excavations, the on-site cohesive soils may be difficult to compact; therefore, a clean granular material may be required in these areas.

Scarified subgrade soils and all fill material should be within 3 percent of the optimum moisture content to facilitate compaction. Furthermore, fill material should not be frozen or placed on a frozen base. It is recommended that all earthwork and site preparation activities be conducted under adequate specifications and properly monitored in the field by a qualified geotechnical testing firm.

6.3 Foundation Excavations

As mentioned in Section 5.1, foundations used to support the structure should have a detailed footing inspection performed for each foundation. A geotechnical engineer or qualified representative should perform these inspections to verify that the exposed materials are similar to those encountered in the borings, loose granular soils have been suitably modified in-place, and that engineered fill has been properly placed and compacted such that it is capable of supporting the design bearing pressure.

We recommend that the foundation excavations be concreted as soon as practical after they are excavated and that water not be allowed to pond in any excavation. If it is necessary to leave the bearing surface open for any extended period of time, we recommend that a thin mat of lean concrete be placed over the bottom of the excavation to reduce damage to the surface from weather or construction. Foundation concrete should not be placed on frozen or saturated subgrade.

Additional foundation subgrade inspection and preparation recommendations are provided in Section 5.1.

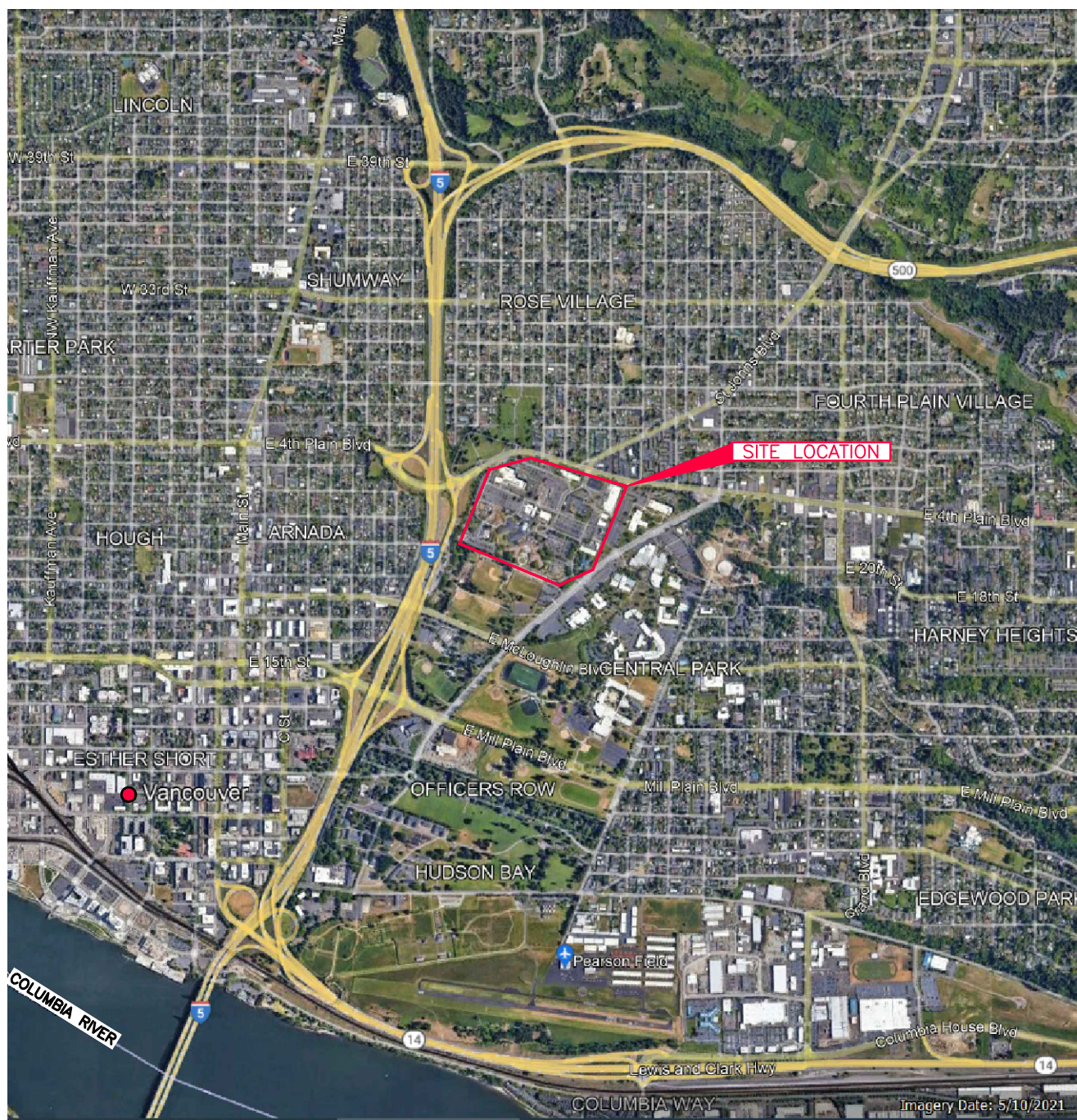
7.0 QUALIFICATION OF RECOMMENDATIONS

Our evaluation of foundation design and construction conditions has been based on our understanding of the site and project information and the data obtained during our field investigation. The general subsurface conditions were based on interpretation of the subsurface data at specific boring locations. Regardless of the thoroughness of a subsurface investigation, there is the possibility that conditions between borings will differ from those at the boring locations, that conditions are not as anticipated by the designers, or that the construction process has altered the soil conditions. The potential for differing conditions is increased at previously developed sites. Therefore, experienced geotechnical engineers should observe earthwork and foundation construction to confirm that the conditions anticipated in design are noted. Otherwise, TTL assumes no responsibility for construction compliance with the design concepts, specifications, or recommendations.

The design recommendations in this report have been developed on the basis of the previously described project characteristics and subsurface conditions. If project criteria or locations change, a qualified geotechnical engineer should be permitted to determine whether the recommendations must be modified. The findings of such a review will be presented in a supplemental report.

The nature and extent of variations between the borings may not become evident until the course of construction. If such variations are encountered, it will be necessary to reevaluate the recommendations of this report after on-site observations of the conditions.

Our professional services have been performed, our findings derived, and our recommendations prepared in accordance with generally accepted geotechnical engineering principles and practices. This warranty is in lieu of all other warranties either expressed or implied. TTL is not responsible for the conclusions, opinions, or recommendations of others based on this data.



LEGEND

— APPROXIMATE SITE LOCATION



APPROXIMATE SCALE - FEET

0 2,000 4,000

PLATE 1.0 SITE LOCATION MAP

PROPOSED CAMPUS SECURITY FENCE
PORTLAND VA MEDICAL CENTER - VANCOUVER CAMPUS
VANCOUVER, WASHINGTON

PREPARED FOR
BLUE TRIDENT, LLC
BAINBRIDGE ISLAND, WASHINGTON

DRAWN TRR/10-19-22

CHECKED CPI/10-24-22

REVISED

APPROVED

JOB NO. 2213101

DRAWING NUMBER

2213101-01G





GOOGLE EARTH IMAGE DATED MAY 10, 2021.

LEGEND

B-1  APPROXIMATE TEST BORING LOCATION

APPROXIMATE SCALE – FEET

0 200 400

PLATE 2.0
TEST BORING LOCATION PLAN
PROPOSED CAMPUS SECURITY FENCE
PORTLAND VA MEDICAL CENTER – VANCOUVER CAMPUS
VANCOUVER, WASHINGTON

PREPARED FOR
BLUE TRIDENT, LLC
BAINBRIDGE ISLAND, WASHINGTON

DRAWN	TRR/10-19-22	CHECKED	CPI/10-24-22
REVISED		APPROVED	
JOB NO.	2213101	 Environmental, Geotechnical Engineering & Testing	
DRAWING NUMBER	2213101-02G		



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BORING NUMBER B-1

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CLIENT <u>Blue Trident, LLC</u>	PROJECT NAME <u>Proposed Campus Security Fence</u>
PROJECT NUMBER <u>2213101</u>	PROJECT LOCATION <u>Portland VAMC Vancouver Campus, Vancouver, WA</u>
DRILLING CONTRACTOR <u>Cascade Drilling Caleb Michael</u>	RIG NO. <u>3230 DT</u> GROUND ELEVATION <u>209 ft</u>
DRILLING METHOD <u>4-Inch Casing</u>	GROUND WATER LEVELS:
DATE STARTED <u>9/22/22</u> COMPLETED <u>9/22/22</u>	▽ AT TIME OF DRILLING <u>15.0 ft / Elev 194.0 ft</u>
LOGGED BY <u>KKC</u> CHECKED BY <u>CPI</u>	AT END OF DRILLING <u>None</u>
NOTES _____	0hrs AFTER DRILLING <u>Backfilled w/Cuttings and Bentonite Chips</u>

ELEVATION (ft)	DEPTH (ft)	GRAPHIC LOG	MATERIAL DESCRIPTION	SAMPLE TYPE NUMBER	RECOVERY % (RQD)	BLOW COUNTS (N VALUE)	UNCONF. COMP. STR. (tsf)	DRY UNIT WT. (pcf)	PL MC LL 20 40 60 80 ▲ SPT N VALUE ▲ 20 40 60 80
	0		TOPSOIL - 3 Inches						
			Moist Loose Brown SILTY SAND w/Gravel (SM)						
205				SS 1	100	4-2-4 (6)	NP		14
	5		Moist Loose Brown POORLY GRADED SAND w/Trace Gravel and Silt (SP)	SS 2	78	5-3-2 (5)	NP		11
200				SS 3	78	5-3-4 (7)	NP		11
	10		Moist Medium Dense Brown POORLY GRADED SAND w/Trace Gravel and Silt (SP)	SS 4	78	5-6-8 (14)	NP		10
195									
	15		Moist Loose Brown POORLY GRADED SAND w/Gravel and Trace Silt (SP)	SS 5	78	5-5-5 (10)	NP		14
			Bottom of hole at 16.5 feet.						

TTL_GEOTECH_STANDARD 2213101.GPJ GINT US LAB.GDT 10/24/22



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CLIENT Blue Trident, LLC

PROJECT NAME Proposed Campus Security Fence

PROJECT NUMBER 2213101

PROJECT LOCATION Portland VAMC Vancouver Campus, Vancouver, WA

DRILLING CONTRACTOR Cascade Drilling Caleb Michael

RIG NO. 3230 DT

GROUND ELEVATION 211 ft

DRILLING METHOD 4-Inch Casing

GROUND WATER LEVELS:

DATE STARTED 9/23/22

COMPLETED 9/23/22

AT TIME OF DRILLING None

LOGGED BY KKC

CHECKED BY CPI

AT END OF DRILLING None

NOTES

0hrs AFTER DRILLING Backfilled w/Cuttings and Bentonite Chips

ELEVATION (ft)	DEPTH (ft)	GRAPHIC LOG	MATERIAL DESCRIPTION	SAMPLE TYPE NUMBER	RECOVERY % (RQD)	BLOW COUNTS (N VALUE)	UNCONF. COMP. STR. (tsf)	DRY UNIT WT. (pcf)	PL 20	MC 40	LL 80	▲ SPT N VALUE ▲
	0											
210			TOPSOIL - 3 Inches Moist Loose Brown SILTY SAND w/Trace Gravel (SM)									
			0.3'									
	5											
205			Moist Medium Dense Brown POORLY GRADED SAND w/Trace Gravel and Silt (SP)	SS 1	100	9-4-4 (8)	NP					11
			5.0'									
	10											
200			@10': pH = 6.8	SS 2	100	8-8-9 (17)	NP					8
				SS 3	100	10-7-7 (14)	NP					7
	15											
195				SS 4	100	10-6-5 (11)	NP					10
	16.5			SS 5	100	8-7-7 (14)	NP					9
			Bottom of hole at 16.5 feet.									



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CLIENT Blue Trident, LLC

PROJECT NAME Proposed Campus Security Fence

PROJECT NUMBER 2213101

PROJECT LOCATION Portland VAMC Vancouver Campus, Vancouver, WA

DRILLING CONTRACTOR Cascade Drilling Caleb Michael

RIG NO. 3230 DT

GROUND ELEVATION 178 ft

DRILLING METHOD 4-Inch Casing

GROUND WATER LEVELS:

DATE STARTED 9/22/22

COMPLETED 9/22/22

AT TIME OF DRILLING None

LOGGED BY KKC

CHECKED BY CPI

AT END OF DRILLING None

NOTES

0hrs AFTER DRILLING Backfilled w/Cuttings and Bentonite Chips

ELEVATION (ft)	DEPTH (ft)	GRAPHIC LOG	MATERIAL DESCRIPTION	SAMPLE TYPE NUMBER	RECOVERY % (RQD)	BLOW COUNTS (N VALUE)	UNCONF. COMP. STR. (tsf)	DRY UNIT WT. (pcf)	PL 20 MC 40 LL 80 ▲ SPT N VALUE ▲
	0		TOPSOIL - 3 Inches Moist Medium Dense Brown SILTY SAND w/Gravel (SM)						
175	5		@5': pH = 7.2	SS 1	78	23-8-9 (17)	NP		8
				SS 2	50	9-8-10 (18)	NP		8
170			Moist Loose Brown SILTY SAND w/Gravel (SM)	SS 3	100	8-4-5 (9)	NP		8
	10		Moist Medium Dense Brown SILTY SAND w/Gravel (SM)	SS 4	100	10-7-7 (14)	NP		7
165	15		Moist Medium Dense Brown POORLY GRADED SAND w/Trace Gravel and Silt (SP)	SS 5	100	13-12-12 (24)	NP		11
	16.5		Bottom of hole at 16.5 feet.						

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CLIENT Blue Trident, LLC

PROJECT NAME Proposed Campus Security Fence

PROJECT NUMBER 2213101

PROJECT LOCATION Portland VAMC Vancouver Campus, Vancouver, WA

DRILLING CONTRACTOR Cascade Drilling Caleb Michael

RIG NO. 3230 DT

GROUND ELEVATION 172 ft

DRILLING METHOD 4-Inch Casing

GROUND WATER LEVELS:

DATE STARTED 9/22/22

COMPLETED 9/22/22

AT TIME OF DRILLING None

LOGGED BY KKC

CHECKED BY CPI

AT END OF DRILLING None

NOTES

0hrs AFTER DRILLING Backfilled w/Cuttings and Bentonite Chips

ELEVATION (ft)	DEPTH (ft)	GRAPHIC LOG	MATERIAL DESCRIPTION	SAMPLE TYPE NUMBER	RECOVERY % (RQD)	BLOW COUNTS (N VALUE)	UNCONF. COMP. STR. (tsf)	DRY UNIT WT. (pcf)	PL 20 MC 40 LL 80 ▲ SPT N VALUE ▲
	0		TOPSOIL - 3 Inches						
			Moist Very Loose Brown SILTY SAND w/Trace Gravel (SM)						
170				SS 1	100	2-2-2 (4)	NP		20
	5		Moist Loose Brown POORLY GRADED SAND w/Trace Gravel and Silt (SP)	SS 2	100	8-4-4 (8)	NP		8
165			Moist Medium Dense Brown SILTY SAND w/Gravel (SM)	SS 3	100	7-7-5 (12)	NP		7
	10		Moist Loose Brown SILTY SAND w/Gravel (SM)	SS 4	100	12-4-4 (8)	NP		7
160									
	15			SS 5	100	7-5-5 (10)	NP		7
	16.5'		Bottom of hole at 16.5 feet.						



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CLIENT Blue Trident, LLC

PROJECT NAME Proposed Campus Security Fence

PROJECT NUMBER 2213101

PROJECT LOCATION Portland VAMC Vancouver Campus, Vancouver, WA

DRILLING CONTRACTOR Cascade Drilling Caleb Michael

RIG NO. 3230 DT

GROUND ELEVATION 171 ft

DRILLING METHOD 4-Inch Casing

GROUND WATER LEVELS:

DATE STARTED 9/23/22 COMPLETED 9/23/22

AT TIME OF DRILLING None

LOGGED BY KKC CHECKED BY CPI

AT END OF DRILLING None

NOTES

0hrs AFTER DRILLING Backfilled w/Cuttings and Bentonite Chips

ELEVATION (ft)	DEPTH (ft)	GRAPHIC LOG	MATERIAL DESCRIPTION	SAMPLE TYPE NUMBER	RECOVERY % (RQD)	BLOW COUNTS (N VALUE)	UNCONF. COMP. STR. (tsf)	DRY UNIT WT. (pcf)	PL 20 MC 40 LL 80 ▲ SPT N VALUE ▲
170	0		Moist Very Stiff Brown SANDY SILT w/Trace Gravel (ML)						
			@2.5': pH = 7.4						
			4.0'	SS 1	100	10-10-13 (23)	NI		10
	5		Moist Medium Dense Brown SILTY SAND w/Trace Gravel (SM)						
			7.5'	SS 2	50	9-8-5 (13)	NP		9
			Moist Loose Brown SILTY SAND w/Trace Gravel (SM)						
			10'	SS 3	78	10-4-3 (7)	NP		6
				SS 4	78	10-4-3 (7)	NP		6
	15		15.0'						
			Moist Medium Dense Brown SILTY SAND w/Gravel (SM)						
			16.5'	SS 5	100	12-9-8 (17)	NP		6
			Bottom of hole at 16.5 feet.						



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CLIENT <u>Blue Trident, LLC</u>	PROJECT NAME <u>Proposed Campus Security Fence</u>
PROJECT NUMBER <u>2213101</u>	PROJECT LOCATION <u>Portland VAMC Vancouver Campus, Vancouver, WA</u>
DRILLING CONTRACTOR <u>Cascade Drilling Caleb Michael</u>	RIG NO. <u>3230 DT</u> GROUND ELEVATION <u>185 ft</u>
DRILLING METHOD <u>4-Inch Casing</u>	GROUND WATER LEVELS:
DATE STARTED <u>9/23/22</u> COMPLETED <u>9/23/22</u>	AT TIME OF DRILLING <u>None</u>
LOGGED BY <u>KKC</u> CHECKED BY <u>CPI</u>	AT END OF DRILLING <u>None</u>
NOTES _____	0hrs AFTER DRILLING <u>Backfilled w/Cuttings, Chips, and Patch</u>

ELEVATION (ft)	DEPTH (ft)	GRAPHIC LOG	MATERIAL DESCRIPTION	SAMPLE TYPE NUMBER	RECOVERY % (RQD)	BLOW COUNTS (N VALUE)	UNCONF. COMP. STR. (tsf)	DRY UNIT WT. (pcf)	PL MC LL 20 40 60 80 ▲ SPT N VALUE ▲ 20 40 60 80
185	0		ASPHALT - 6 Inches						
			Moist Brown SILTY SAND w/Trace Gravel (SM)						
			0.5'						
			2.5'						
			Moist Stiff Brown SANDY SILT w/Trace Gravel and Organics (ML) (Organic Content = 3.6%)	SS 1	100	10-8-6 (14)	NI		12
180	5		5.0'						
			Moist Loose Brown POORLY GRADED SAND w/Trace Gravel and Silt (SP)	SS 2	100	12-7-3 (10)	NP		10
			@7.5': w/Gravel	SS 3	78	7-4-3 (7)	NP		10
			9.0'						
			Moist Loose Brown SILTY SAND w/Trace Gravel (SM)						
175	10		10.0'						
			Moist Medium Dense Brown POORLY GRADED SAND w/Trace Gravel and Silt (SP)	SS 4	78	9-7-7 (14)	NP		6
			15.0'						
170	15		Moist Loose Brown POORLY GRADED SAND w/Gravel and Trace Silt (SP)	SS 5	78	7-4-4 (8)	NP		6
			16.5'						
			Bottom of hole at 16.5 feet.						

TTL_GEOTECH_STANDARD 2213101.GPJ GINT US LAB.GDT 10/24/22



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CLIENT	Blue Trident, LLC	PROJECT NAME	Proposed Campus Security Fence
PROJECT NUMBER	2213101	PROJECT LOCATION	Portland VAMC Vancouver Campus, Vancouver, WA
DRILLING CONTRACTOR	Cascade Drilling Caleb Michael	RIG NO.	3230 DT
DRILLING METHOD	4-Inch Casing	GROUND ELEVATION	204 ft
DATE STARTED	9/23/22	COMPLETED	9/23/22
LOGGED BY	KKC	CHECKED BY	CPI
NOTES			
GROUND WATER LEVELS:		AT TIME OF DRILLING	
		None	
		AT END OF DRILLING	
		None	
		0hrs AFTER DRILLING	
		Backfilled w/Cuttings and Bentonite Chips	

ELEVATION (ft)	DEPTH (ft)	GRAPHIC LOG	MATERIAL DESCRIPTION	SAMPLE TYPE NUMBER	RECOVERY % (RQD)	BLOW COUNTS (N VALUE)	UNCONF. COMP. STR. (tsf)	DRY UNIT WT. (pcf)	PL 20 MC 40 LL 80 ▲ SPT N VALUE ▲
	0		TOPSOIL - 3 Inches						
			Moist Loose Brown POORLY GRADED SAND w/Silt and Gravel (SP-SM) (pH = 6.9)						
200			0.3'	SS 1	100	5-5-4 (9)	NP		7
	5		5.0'	SS 2	100	11-11-11 (22)	NP		6
			7.5'	SS 3	100	12-12-13 (25)	NP		7
195				SS 4	100	11-11-7 (18)	NP		7
	10								
				SS 5	100	8-6-7 (13)	NP		12
190									
	15								
			16.5'						
			Bottom of hole at 16.5 feet.						

TTL_GEOTECH_STANDARD 2213101.GPJ GINT US LAB.GDT 10/24/22

LEGEND KEY

Unified Soil Classification System Soil Symbols

	GW - WELL GRADED GRAVEL Includes Gravel-Sand mixtures, little or no fines.		GP - POORLY GRADED GRAVEL Includes Gravel-Sand mixtures, little or no fines.		GM - SILTY GRAVEL Includes Gravel-Sand-Silt mixtures.		GC - CLAYEY GRAVEL Includes Gravel-Sand-Clay mixtures.
	SW - WELL GRADED SAND Includes Gravelly Sands, little or no fines.		SP - POORLY GRADED SAND Includes Gravelly Sands, little or no fines.		SM - SILTY SAND Includes Sand-Silt mixtures.		SC - CLAYEY SAND Includes Sand-Clay mixtures.
	ML - SILT Includes Silt with Sand and Sandy Silt.		CL - LEAN CLAY Includes Sandy Lean Clay and Lean Clay with Sand and Gravel.		MH - ELASTIC SILT Includes Sandy Elastic Silt and Elastic Silt with Sand.		CH - FAT CLAY Includes Sandy Fat Clay and Fat Clay with Sand.
	CL-ML - SILTY CLAY Includes Clayey Silt of low plasticity.		OL - ORGANIC SILT and ORGANIC CLAY of low plasticity.		OH - ORGANIC SILT and ORGANIC CLAY of medium to high plasticity.		Pt - PEAT Includes humus, swamp and other soils with high organic content.
	FILL MATERIAL - Includes controlled and non-controlled soil and non-soil materials.		TOPSOIL		ASPHALT - Bituminous Asphalt		CONCRETE - Includes broken concrete rubble.

Sample Symbols

	SS - Split Spoon		ST - Shelby Tube		RC - Rock Core		GS - Geoprobe Sleeve
			AU - Auger Cuttings		GB - Grab		

Notes:

1. Exploratory borings were performed on September 22 and 23, 2022. The borings were advanced using 4-inch diameter casing.
2. These logs are subject to the limitations, conclusions, and recommendations in the report and should not be interpreted separate from the report.
3. The test borings were located in the field by Cascade based on a proposed plan that was approved by Mr. Andrew S. Thornton of Portland VA Medical Center, Vancouver Campus on August 10, 2022. Ground surface elevations at the boring locations were estimated using Google Earth.
4. Unconfined Compressive Strength (tsf):
NP = Non-Plastic
NI = Non-Intact



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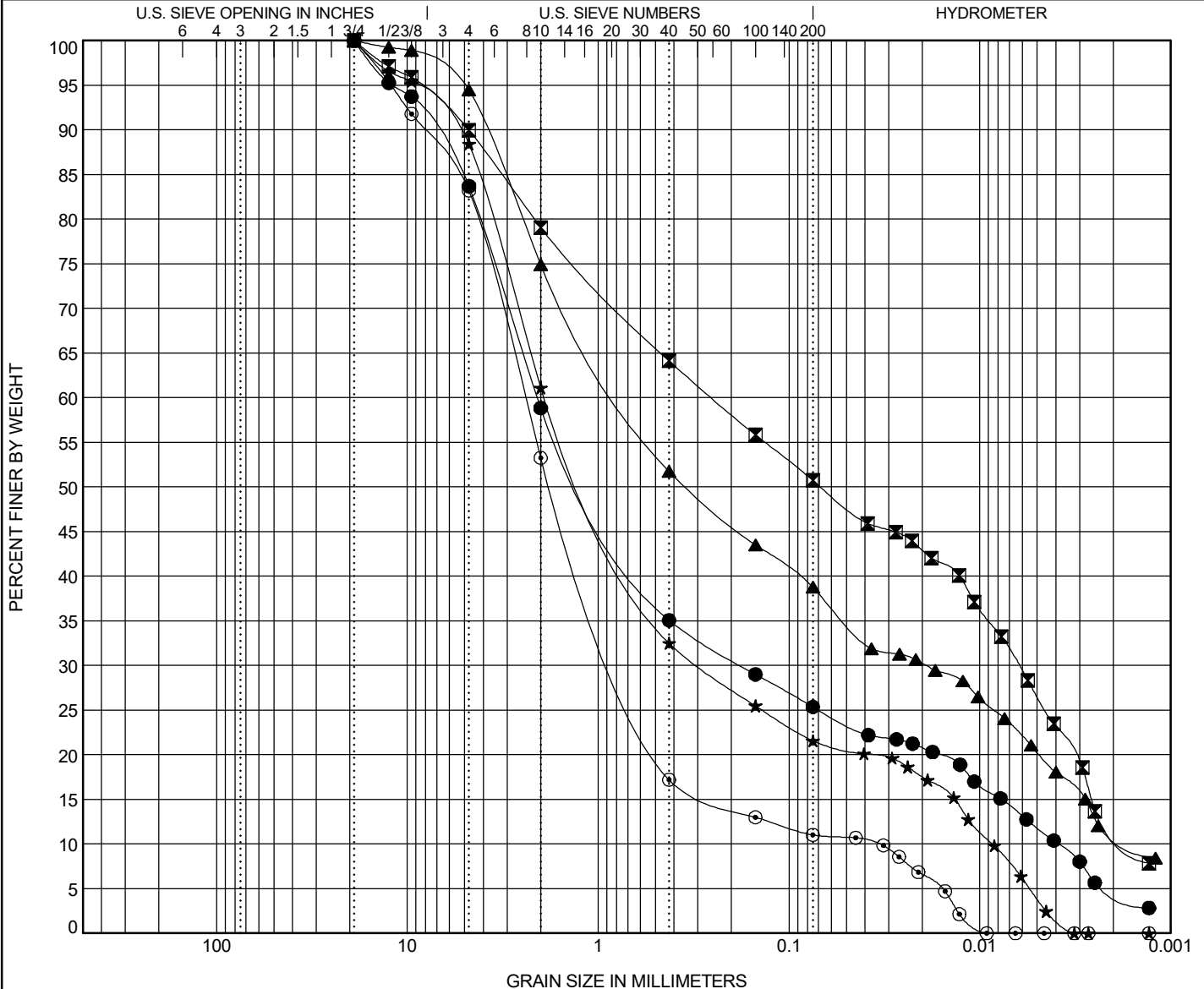
GRAIN SIZE DISTRIBUTION

CLIENT Blue Trident, LLC

PROJECT NAME Proposed Campus Security Fence

PROJECT NUMBER 2213101

PROJECT LOCATION Portland VAMC Vancouver Campus, Vancouver, WA



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

Specimen Identification		USCS Classification				LL	PL	PI	Cc	Cu
● B-1	2.5	SILTY SAND with GRAVEL (SM)				34	26	8	3.9	534.9
▣ B-3	2.5	SANDY SILT (ML)				35	26	9	0.1	152.7
▲ B-5	2.5	SILTY SAND (SM)				34	25	9	0.3	454.5
★ B-6	10.0	SILTY SAND (SM)				NP	NP	NP	5.3	219.9
⊙ B-8	10.0	POORLY GRADED SAND with SILT and GRAVEL (SP-SM)				NP	NP	NP	6.6	71.6

Specimen Identification		D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay
● B-1	2.5	19	2.082	0.178	0.004	16.3	58.3	13.5	11.8
▣ B-3	2.5	19	0.253	0.006	0.002	10.0	39.2	24.2	26.6
▲ B-5	2.5	19	0.738	0.019	0.002	5.5	55.7	18.5	20.3
★ B-6	10.0	19	1.884	0.293	0.009	11.6	66.9	17.8	3.8
⊙ B-8	10.0	19	2.43	0.736	0.034	16.8	72.2	11.0	0.0

GRAIN SIZE 2213101.GPJ GINT US LAB.GDT 10/24/22